

REPRESENTATIVE OUTPUT SAMPLES ILLUSTRATING THE PERFORMANCE SPECTRUM OF LLMs

The following representative outputs were selected from a set of seventy open-ended problems involving applications of first-order Ordinary Differential Equations. They are presented to illustrate the strengths and limitations of Large Language Models with respect to mathematical modeling, Python code generation and solution accuracy. The results reveal notable variations in performance across models, reflecting differences in their mathematical reasoning capabilities and problem-solving effectiveness.

Question ID:32.

A body whose temperature is 180° is immersed in a liquid which is kept at a constant temperature of 60° . In one minute, the temperature of the immersed body decreases to 120° . How long will it take for the body's temperature to decrease to 90° .

Ground Truth Solution:

Using Newton's Law of Cooling, we have the formula: $T(t) = T_{env} + (T_0 - T_{env})e^{-kt}$ Given: $T_{env} = 60^\circ$, $T_0 = 180^\circ$, $T(1) = 120^\circ$ We substitute into the formula: $T(t) = 60 + (180 - 60)e^{-kt} = 60 + 120e^{-kt}$ At $t = 1$, $T(1) = 120$, so: $120 = 60 + 120e^{-k}$ $\rightarrow 60 = 120e^{-k}$ $\rightarrow 1/2 = e^{-k}$ Take the natural logarithm of both sides: $-\ln(2) = -k$ $\rightarrow k = \ln(2) \approx 0.6931$ Now, we want to find t such that $T(t) = 90$. Using the equation: $90 = 60 + 120e^{-k t}$ $\rightarrow 30 = 120e^{-k t}$ $\rightarrow 1/4 = e^{-k t}$ Take the natural logarithm: $\ln(1/4) = -0.6931 t$ $\rightarrow -\ln(4) = -0.6931 t$ $\rightarrow t = \ln(4)/0.6931 \approx 1.3863/0.6931 \approx 2$ Therefore, it will take approximately 2 minutes for the body's temperature to decrease to 90° .

Ground Truth Answer: 2

Response: *inception/mercury-2_class-Three shot prompt*

Tasku202f01 – Modeling Let $T(t)$ be the temperature of the body at time t (minutes). The surrounding liquid has constant temperature $T_s = 60^\circ$. According to Newton's law of cooling the rate of temperature change is proportional to the temperature difference: $\frac{dT}{dt} = k(T_s - T)$ where $k > 0$ is the cooling constant. This is a linear first-order ODE. Solving it gives $T(t) = T_s + (T_0 - T_s)e^{-kt}$ where $T_0 = T(0)$ is the initial temperature. Initial condition: $T(0) = 180^\circ$. After one minute the temperature is $T(1) = 120^\circ$. Substituting these values: $120 = 60 + (180 - 60)e^{-k}$ $\rightarrow 60 = 120e^{-k}$ $\rightarrow \ln(1/2) = -k$ $\rightarrow k = \ln(2)$. To find the time when the temperature reaches 90° set $T(t) = 90$: $90 = 60 + 120e^{-kt}$ $\rightarrow 30 = 120e^{-kt}$ $\rightarrow \ln(1/4) = -kt$ $\rightarrow t = \frac{\ln(4)}{k} = \frac{\ln(4)}{\ln(2)} = 2$ minutes.

Tasku202f02 – Python code pythonimport numpy as npfrom scipy.integrate import solve_ivpndef model(t, T, k):
 return [k * (60 - T)]
nk = np.log(0.5)
nsol = solve_ivp(model, [0, 10], [180], args=(k,), dense_output=True)nt_vals = np.linspace(0, 10, 10001)
T_vals = sol.sol(t_vals)[0]
idx = np.argmax(np.abs(T_vals - 90))
nresult = round(t_vals[idx], 5)
nprint("<result>", result,
"</result>")

Answer: 1.999 ✓

Response: *openai/gpt-5.4_class-Three shot prompt*

Solution 1: Modeling the problem using an ODE Let $T(t)$ be the temperature of the body at time t minutes. Since the body is placed in a liquid with constant temperature 60 degrees, Newton's law of cooling gives the differential equation $\frac{dT}{dt} = k(60 - T)$ where k is a positive constant. Step 1: Solve the differential equation. The standard solution of $\frac{dT}{dt} = k(60 - T)$ is $T(t) = 60 + (T_0 - 60)e^{-kt}$. Since the initial temperature is $T(0) = 180$, we get $T(t) = 60 + (180 - 60)e^{-kt}$. Step 2: Use the condition at $t = 1$ minute. After 1 minute, the temperature is 120 , so $120 = 60 + 120e^{-k}$. Subtract 60 from both sides: $60 = 120e^{-k}$. Divide by 120 : $e^{-k} = 1/2$. Take the natural logarithm: $-\ln(2) = -k$ $\rightarrow k = \ln(2)$. Step 3: Find the time when the temperature reaches 90 degrees. Set $T(t) = 90$: $90 = 60 + 120e^{-kt}$. Subtract 60 : $30 = 120e^{-kt}$. Divide by 120 : $e^{-kt} = 1/4$. Take the natural logarithm: $-\ln(4) = -kt$ $\rightarrow t = \frac{\ln(4)}{k} = \frac{\ln(4)}{\ln(2)} = 2$ minutes. Therefore, the body will take 2 minutes to decrease to 90 degrees.

pythonimport numpy as npfrom scipy.integrate import solve_ivpndef model(t, T, k):
 return [k * (60 - T)]
nk = np.log(2)
nsol = solve_ivp(model, [0, 2], [180], args=(k,), t_eval=[2])
nresult = round(sol.y[0][-1], 5)
nprint(result)

Answer: 15.915915915917 ✗

Figure 1: llm generated ODE model and python code for newtons law of cooling problem:mercury-2 vs gpt-5.4

Question ID:68.
Determine the interest rate required to double an investment in eight years under continuous compounding?

Ground Truth Solution:
Let $A(t)$ denote the investment at a time t . The ODE is given by $dA/dt = kA$ and solving this equation using integrating factor we have $A(t) = Ce^{kt}$. Since we are not given the initial condition, we assume that at $t = 0$, $A_0 = Ce^{k(0)} \Rightarrow C = A_0$. Hence $A(t) = A_0e^{kt}$. We want to find k (interest rate) when we have $2A_0$ and $t = 8$. Therefore $2A_0 = A_0e^{8k} \Rightarrow e^{8k} = 2 \Rightarrow 8k = \ln 2 \Rightarrow k = (\ln 2)/8 \approx 0.0866$. Thus the interest rate is given by $k = (\ln 2)/8 \approx 0.0866$

Ground Truth Answer: 0.0866

Response: *openai/gpt-4.1-Three-shot prompt*
`"""Solution 1: Modeling the problem using an ODE"""
Let A(t) be the amount of money at time t (in years). For continuous compounding, the rate of change of the amount is proportional to the current amount: dA/dt = r * A
where r is the interest rate.
The solution to this ODE is: A(t) = A0 * exp(r * t)
where A0 is the initial amount.
We are told that the investment doubles in 8 years, so: A(8) = 2 * A0
Substituting into the solution: 2 * A0 = A0 * exp(r * 8)
Dividing both sides by A0: 2 = exp(r * 8)
Taking the natural logarithm of both sides: ln(2) = r * 8
Solving for r: r = ln(2) / 8
Therefore, the required interest rate r is equal to the natural logarithm of 2 divided by 8.
"""
Solution 2: Convert the ODE model into Python code
python
import numpy as np
r = np.log(2) / 8
result = round(float(r), 5)
print("<result>", result, "</result>")
"""
Answer: 0.08664 ✓`

Response: *google/gemma-2-27b-it-Three-shot prompt*
`"""Solution 1: Modeling the problem using an ODE"""
Let A(t) be the amount of money at time t. The rate of change of the investment is proportional to the current amount: dA/dt = kA
where k is the interest rate.
Solving this differential equation, we obtain A(t) = Ae^{kt}
Applying the initial condition A(0) = A_0, we obtain A = A_0. Hence A(t) = A_0e^{kt}
Now, we want to find the interest rate k that doubles the investment in 8 years, meaning A(8) = 2A_0. Substituting this into the equation: 2A_0 = A_0e^{8k}
Dividing both sides by A_0: 2 = e^{8k}
Taking the natural logarithm of both sides: ln(2) = 8k
Solving for k: k = ln(2) / 8
Therefore, the interest rate required to double the investment in eight years under continuous compounding is k = ln(2) / 8.
"""
Solution 02: Convert the ODE model into Python code
python
import numpy as np
from scipy.integrate import solve_ivp
def model(t, A, k):
 return k * A
nk = np.log(2) / 8
sol = solve_ivp(model, [0, 8], [1], args=(k,), t_eval=[8])
result = round(sol.y[0][0], 5)
print("<result>", result, "</result>")
"""
Answer: 2.0 ✗`

Figure 2: llm generated ODE model and python code for economy problem:gpt-4.1 vs gemma -2-27b-it

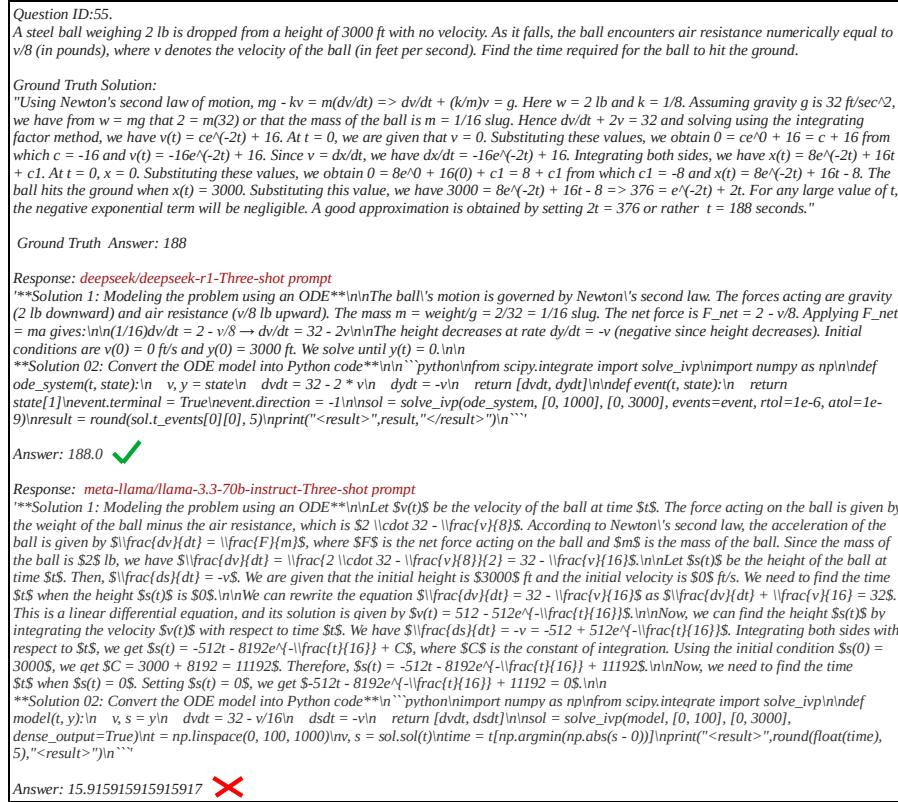


Figure 3: llm generated ODE model and python code for falling object problem:deepseek-r-1 vs llama3-3-70b-instruct

Question ID:01.
A tank contains 1000 L of salt water with 15 kg of dissolved salt. Pure water enters the tank at a rate of 10 L/min. The solution is kept well-mixed and drains from the tank at the same rate. Calculate how much salt remains in the tank after 20 minutes.

Ground Truth Solution:
"Let $S(t)$ be the amount of salt (in kilograms) in the tank at time t (in minutes). Since the inflow and outflow rates are equal, the volume in the tank remains constant at 1000 liters. We are given that: Inflow of pure water: no salt is added; Outflow: the concentration of salt at time t is $S(t)/1000$ kg/L; Outflow rate of salt: $10 * S(t)/1000 = 1/100 S(t)$ kg/min. Thus, the rate of change of salt in the tank is: $dS/dt = -1/100 S(t) \rightarrow dS/S = -1/100 dt$ Integrate both sides: $\int 1/S dS = \int -1/100 dt \rightarrow \ln |S| = -t/100 + C \rightarrow S(t) = Ce^{-(t/100)}$ Applying the initial condition $S(0) = 15$, we have: $S(0) = C * e^0 = C \rightarrow C = 15$ Hence, $S(t) = 15 e^{-(t/100)}$ Calculating the amount of salt at $t = 20$, we have $S(20) = 15 e^{-(20/100)} = 15 e^{-(0.2)}$ Using the approximation $e^{-(0.2)} \approx 0.8187$: $S(20) \approx 15 * 0.8187 = 12.28$ kg. Therefore, after 20 minutes, we have approximately 12.3 kg"

Ground Truth Answer: 12.3

Response: *meta-llama/llama-3.3-70b-instruct-Zero shot prompt*
"## Step 1: Define the problem and identify the key components\nThe problem involves a tank with an initial volume of 1000 L of salt water containing 15 kg of dissolved salt. Pure water enters the tank at a rate of 10 L/min, and the solution drains from the tank at the same rate. We need to calculate the amount of salt remaining in the tank after 20 minutes.\n## Step 2: Formulate the ordinary differential equation (ODE) model\nLet's denote the amount of salt in the tank at time t as $S(t)$ in kg. The rate of change of salt in the tank is given by the difference between the rate at which salt enters the tank and the rate at which it leaves. Since pure water enters the tank, the rate at which salt enters is 0 kg/min. The rate at which salt leaves the tank is given by the concentration of salt in the tank multiplied by the flow rate. The concentration of salt is $S(t) / 1000$ kg/L (since the volume is 1000 L), and the flow rate is 10 L/min. Therefore, the rate at which salt leaves the tank is $(S(t) / 1000) * 10$ kg/min.\n## Step 3: Write down the ODE\nThe ODE representing the rate of change of salt in the tank is $dS/dt = - (S(t) / 1000) * 10$, which simplifies to $dS/dt = -0.01 * S(t)$. \n## Step 4: Define the initial condition\nThe initial amount of salt in the tank is given as 15 kg, so $S(0) = 15$. \n## Step 5: Convert the ODE model into Python code\n```python\nimport numpy as np\nfrom scipy.integrate import odeint\n\ndef model(S, t):\n dSdt = -0.01 * S\n return dSdt\n\nS0 = 15\nnt = np.linspace(0, 20)\nS = odeint(model, S0, nt)\nfinal_salt = round(S[-1][0], 5)\n\nprint(final_salt)\n```\n\nAnswer: 12.28096 ✓

Response: *google/gemma-2-27b-it-Zero shot prompt*
"## Task 01:\n1. Define Variables: Let $S(t)$ represent the amount of salt (in kg) in the tank at time t (in minutes). \n2. Rate of Change: The rate of change of salt in the tank is determined by the difference between the rate of salt entering and the rate of salt leaving. \n3. Salt entering: Since pure water enters, no salt is added. \n4. Salt leaving: The concentration of salt in the tank is $S(t) / 1000$ kg/L. The rate of salt leaving is then $(S(t) / 1000) * 10$ kg/min = $S(t) / 100$ kg/min. \n5. ODE Formulation: The ODE describing the system is: $dS/dt = - S(t) / 100$. \n## Task 02:\npython\nfrom scipy.integrate import solve_ivp\n\ndef dSdt(t, S):\n return -S / 100\n\nS0 = 15\nnt_span = (0, 20)\n\nsol = solve_ivp(dSdt, nt_span, [S0], dense_output=True)\n\nround(sol.y[-1][0], 5)\n\n"

Answer: 15 ✗

Figure 4: llm generated ODE model and python code for mixing problem: llama3-3-70b-instruct vs gemma -2-27b-it

Question ID:51.
The population of the municipality of Sambo, Angola, is known to increase at a rate proportional to the number of people presently living in the municipality. If after two years the population has doubled, and after three years the population is 20000, estimate the number of people initially living in Sambo.

Ground Truth Solution:
Let N denote the number of people living in Sambo, Angola, at any time t and let N_0 denote the number of people initially living in Sambo. We have, $dN/dt = kN = 0$ which has the solution $N = ce^{kt}$. At $t = 0$, $N = N_0$; hence, it follows that $N_0 = ce^{k(0)}$, or that $c = N_0$. Thus, $N = N_0e^{kt}$. At $t = 2$, $N = 2N_0$. Substituting these values, we have $2N_0 = N_0e^{2k}$ from which $k = (1/2)\ln 2 = 0.347$. Substituting this value gives $N = N_0e^{(0.347)t}$. At $t = 3$, $N = 20,000$. Substituting these values, we obtain $20000 = N_0e^{(0.347)(3)} = N_0(2.832)$ or $N_0 = 7062$.

Ground Truth Answer: 7062

Response: *anthropic/claude-haiku-4.5-Zero shot prompt*
Task 01: ODE Model Formulation
Step-by-Step Reasoning
Problem Analysis: We need to model population growth where the rate of change is proportional to the current population.
Setting up the ODE: Let $P(t)$ represent the population at time t (in years). The fundamental principle states: $dP/dt = kP$, where k is the proportionality constant. This is a first-order linear ODE.
General Solution: The solution to $dP/dt = kP$ is: $P(t) = P_0 \cdot e^{(k \cdot t)}$ where P_0 is the initial population at $t=0$.
Applying Given Conditions:
Condition 1: After 2 years, population doubles. $P(2) = 2P_0$.
 $2P_0 = P_0 \cdot e^{(k \cdot 2)} \Rightarrow 2 = e^{2k} \Rightarrow \ln 2 = 2k \Rightarrow k = \ln(2)/2$
Condition 2: After 3 years, population is 20000. $P(3) = 20000$.
 $20000 = P_0 \cdot e^{(k \cdot 3)} = P_0 \cdot e^{(3 \cdot \ln(2)/2)} = P_0 \cdot e^{(1.5 \ln 2)} = P_0 \cdot 2^{1.5} = P_0 \cdot 2\sqrt{2}$
 $P_0 = 20000 / (2\sqrt{2}) \approx 7071.07$
Task 02: Python Code Implementation
python
import numpy as np
from scipy.integrate import odeint
def population_model(P, t, k):
 return k * P
nk = np.log(2) / 2
P0_calculated = 20000 / (2**(3/2))
nk_eval = np.array([0, 3])
solution = odeint(population_model, [P0_calculated], nk_eval, args=(k,))
final_population = solution[-1, 0]
initial_population = P0_calculated
print(f"initial_population: {5f}")
Output: 7071.67811865475
Answer: 7071.67811865475

Response: *google/gemma-2-27b-it-Zero shot prompt*
Task 01: Let $P(t)$ represent the population of Sambo at time t (in years). We are told the population growth rate is proportional to the current population. This translates to the following differential equation: $dP/dt = k \cdot P(t)$ where k is the proportionality constant. This is a separable differential equation. We can solve it by separating the variables and integrating:
$$\frac{dP}{P} = k \, dt$$

Integrating both sides:
$$\int \frac{dP}{P} = \int k \, dt$$

This gives us:
$$\ln|P(t)| = kt + C$$

where C is the constant of integration. We can solve for $P(t)$:
$$P(t) = \exp(kt + C) = A \cdot \exp(kt)$$

where $A = \exp(C)$ is another constant. Now we use the given information to find the values of A and k . We know that $P(2) = 2 \cdot P(0)$. Substituting into our equation:
$$P(2) = A \cdot \exp(2k) = 2 \cdot A \cdot \exp(0) = 2A$$

This means $A \cdot \exp(2k) = 2A$, so $\exp(2k) = 2$. We also know that $P(3) = 20000$. Substituting:
$$P(3) = A \cdot \exp(3k) = 20000$$

We can use the fact that $\exp(2k) = 2$ to solve for A :
$$A = 20000 / \exp(3k)$$

Substituting this back into the equation $\exp(2k) = 2$, we can solve for k . Finally, knowing k and A , we can find the initial population $P(0) = A$.
Task 02: Python Code Implementation
python
from scipy.integrate import solve_ivp
def model(t, P):
 k = 0.69315
 return k * P
nsol = solve_ivp(model, (0, 3), [10000], dense_output=True)
P_0 = sol.y[0][0]
round(P_0, 5)
Answer: 10000

Figure 5: llm generated ODE model and python code for exponential growth problem:haiku4-5 vs gemma -2-27b-it

Question ID:20.
A resistor $R=5\Omega$ and an inductor $L=1H$ are connected in series to a $10V$ DC power supply. If the switch is closed at $t=0$, find the initial current $I(0)$.

Ground Truth Solution:
'KVL gives: $L \frac{dI}{dt} + RI = V \rightarrow \frac{dI}{dt} + 5I = 10$ Integrating factor: e^{5t} Multiply through: $e^{5t} \frac{dI}{dt} + 5e^{5t} I = 10e^{5t} \rightarrow \frac{d}{dt} (I e^{5t}) = 10e^{5t}$ Integrate: $I e^{5t} = 2e^{5t} + C \rightarrow I(t) = 2 + Ce^{5t}$ Initial condition: $I(0) = 0 \rightarrow 0 = 2 + C \rightarrow C = -2$ So: $I(t) = 2 - 2e^{5t} \rightarrow I(0) = 0$

Ground Truth Answer: 0

Generated Response:anthropic/claude-haiku-4.5-Zero shot

Task 01: ODE Model Formulation

Problem Analysis

We have an RL circuit (resistor-inductor circuit) connected to a DC power supply. We need to model the current behavior when the switch closes at $t=0$.

Step-by-Step Reasoning

1. Circuit Configuration:

- Resistor: $R = 5\Omega$
- Inductor: $L = 1H$
- Voltage source: $V = 10V$ (DC)
- Initial condition: Switch closes at $t = 0$

2. Applying Kirchhoff's Voltage Law (KVL):

The sum of voltages around the circuit equals zero:

$$V - V_R - V_L = 0$$

Where:

- $V_R = I \cdot R$ (voltage across resistor)
- $V_L = L \cdot \frac{dI}{dt}$ (voltage across inductor)

3. Formulating the Differential Equation:

$$V = I \cdot R + L \cdot \frac{dI}{dt}$$

Rearranging:

$$L \cdot \frac{dI}{dt} = V - I \cdot R$$

$$\frac{dI}{dt} = \frac{V - I \cdot R}{L}$$

4. Substituting Known Values:

$$\frac{dI}{dt} = \frac{10 - I \cdot 5}{1}$$

5. Initial Condition:

At $t = 0$, the switch just closes. The inductor opposes any instantaneous change in current, so:

$$I(0) = 0$$

6. ODE Model Summary:

Differential Equation: $\frac{dI}{dt} = 10 - 5I$

Initial Condition: $I(0) = 0$

Domain: $t \geq 0$

Task 02: Python Implementation

```
python
from scipy.integrate import odeint
import numpy as np

def rl_circuit(I, t, R, L, V):
    dI_dt = (V - I * R) / L
    return dI_dt

R = 5
L = 1
V = 10
I0 = 0
t0 = 0
t1 = 10
solution = odeint(rl_circuit, I0, [t0, t1], args=(R, L, V))
initial_current = solution[0][0]
print(f"Initial current: {initial_current}")
Output: 0.000000
```

Answer: 0.0 ✓

Generated Response:meta-llama/llama-3.3-70b-instruct-Zero shot

Step 1: Understand the problem and identify the key components

The problem involves a resistor (R) and an inductor (L) connected in series to a DC power supply. The given values are $R = 5\Omega$, $L = 1H$, and the voltage of the power supply is $10V$. We need to find the initial current $I(0)$ when the switch is closed at $t = 0$.

Step 2: Apply Kirchhoff's voltage law to derive the differential equation

According to Kirchhoff's voltage law, the sum of the voltage drops across the resistor and the inductor equals the voltage of the power supply. The voltage drop across the resistor is RI , and the voltage drop across the inductor is $L \frac{dI}{dt}$, where I is the current and $\frac{dI}{dt}$ is the derivative of the current with respect to time. Therefore, the equation becomes:

$$RI + L \frac{dI}{dt} = 10$$

Step 3: Rearrange the equation to isolate $\frac{dI}{dt}$

To solve for the current I , we need to isolate $\frac{dI}{dt}$. Rearranging the equation gives:

$$\frac{dI}{dt} = \frac{10 - RI}{L}$$

Step 4: Identify the initial condition

Since we are looking for the initial current $I(0)$, we need to find the value of I at $t = 0$. At $t = 0$, the current in the circuit is 0 because the switch has just been closed and the inductor has not yet started to conduct.

Step 5: Solve for the initial current $I(0)$

However, realizing that the initial current in an RL circuit when first connected to a voltage source is determined by the resistor since the inductor initially acts like an open circuit, we can directly calculate $I(0)$ using Ohm's law for the resistor. Thus, $I(0) = 10V / 5\Omega = 2A$.

```
python
import numpy as np
from scipy.integrate import solve_ivp

def model(t, I, R, L):
    dIdt = (10 - R * I) / L
    return dIdt

R = 5
L = 1
t0 = 0
t1 = 10
I0 = 0
I_end = 10 / R
sol = solve_ivp(model, [t0, t1], [I0], args=(R, L), t_eval=[t0, t1])
print(round(sol.y[0], 5))
```

Answer: 2.0 ✗

Figure 6: llm generated ODE model and python code for electrical circuit problem:haiku4-5 vs llama3-3-70b-instruct

Question ID:58.
A person places \$20,000 in a savings account which pays 5% interest per annum, compounded continuously. Find the time required for the account to double in value, presuming no withdrawals and no additional deposits.

Ground Truth Solution:
Let $N(t)$ denote the balance in the account at any time t . Initially, $N(0) = 20,000$. The balance grows by the accumulated interest payments, which are proportional to the amount of money in the account. The constant of proportionality is the interest rate. In this case, $k = 0.05$ and hence we have $dN/dt - 0.05N = 0$. This differential equation is both linear and separable and its solution is $N(t) = ce^{(0.05t)}$. At $t = 0$, $N(0) = 20,000$, which yields $20,000 = ce^0 = c$. With this value of c , $N(t) = 20,000e^{(0.05t)}$. We seek the time t at which $N(t) = \$40,000$. Substituting these values and solving for t , we obtain $40,000 = 20,000e^{(0.05t)} \Rightarrow 2 = e^{(0.05t)} \Rightarrow \ln|2| = 0.05t \Rightarrow t = (1/0.05)\ln|2| = 13.86$ years.

Ground Truth Answer: 13.86

Response: *openai/gpt-4.1 class-Three-shot prompt*

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***Solution 1: Modeling the problem using an ODE**\nLet A(t) be the amount in the savings account at time t (in years). The account earns interest at a continuous annual rate of 5%. The rate of change of the amount is proportional to the current amount: \nndA/dt = 0.05 * A\n\nThis is a first-order linear ODE. The general solution is: \nA(t) = A0 * exp(0.05 * t)\n\nwhere A0 is the initial amount. Given A0 = 20,000. \n\nWe are asked to find the time t when the account doubles, that is, when A(t) = 40,000. \n\nSo, \n40,000 = 20,000 * exp(0.05 * t)\n\nDividing both sides by 20,000: \n2 = exp(0.05 * t)\n\nTaking the natural logarithm of both sides: \nln(2) = 0.05 * t\n\nSolving for t: \nt = ln(2) / 0.05\n\nTherefore, the time required for the account to double is t = ln(2) / 0.05 years. \n\n---\n\n**Solution 2: Convert the ODE model into Python code**\n\npython\nimport numpy as np\n\nresult = round(np.log(2) / 0.05, 5)\n\nprint("<result>".result,"</result>")\n

```

Answer: 13.86294 ✓

Response: *google/gemma-2-27b-it- Three-shot prompt*

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## Solution 1: Modeling the problem using an ODE\nLet A(t)$ be the amount in the account at time $t$. The rate of change of the amount is proportional to the current amount: \nndA/dt = kA\n\nwhere $k$ is the interest rate. \n\nSolving this differential equation, we obtain: \nA(t) = Ae^{(kt)}\n\nwhere A is the initial amount. \n\nWe are given that A(0) = 20000, so A = 20000. Hence: \nA(t) = 20000e^{(kt)}\n\nWe want to find the time $t$ when A(t) = 40000 (double the initial amount). Setting A(t) = 40000, we get: \n40000 = 20000e^{(kt)}\n\nDividing both sides by 20000: \n2 = e^{(kt)}\n\nTaking the natural logarithm of both sides: \nln(2) = kt\n\nSolving for t: \nt = ln(2) / k\n\nSince the interest rate is 5% per annum, k = 0.05. Therefore: \nt = ln(2) / 0.05\n\nCalculating the value: \nt \u2248 13.86\n\nTherefore, it will take approximately **13.86 years** for the account to double in value. \n\n\n## Solution 02: Convert the ODE model into Python code\n\npython\nfrom scipy.integrate import solve_ivp\nimport numpy as np\n\ndef model(t, A):\n    return 0.05 * A\n\nsol = solve_ivp(model, [0, 20], [20000], t_eval=[20])\nresult = round(sol.y[0][0], 5)\n\nprint("<result>".result,"</result>")\n

```

Answer: 54366.53974 ✗

Figure 7: llm generated ODE model and python code for economy problem:gpt-4.1 vs gemma -2-27b-it